Elementary Problems
Arithmetic Function and Floor Function
Solutions

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Divisor Function

1. Since $20 = 2^2 \times 5$ and $36 = 2^2 \times 3^2$, we have

$$\begin{aligned} d(20) &= (2+1)(1+1) = 6, \\ d(36) &= (2+1)(2+1) = 9. \end{aligned}$$

2. Since $30 = 2 \times 3 \times 5$ and $72 = 2^3 \times 3^2$, we have

$$\begin{aligned} \sigma(30) &= (1+2)(1+3)(1+5) = 72, \\ \sigma(72) &= (1+2+2^2+2^3)(1+3+3^2) = 195. \end{aligned}$$

3. Note that $n = 1$ is a solution. Suppose $n \geqslant 2$. If $d(n) = n$, then each of $1, 2, \ldots, n$ is a divisor of n. In particular, we have $n - 1 \mid n$. This yields $n - 1 \mid 1$. The only solution in this case is $n = 2$.

   Therefore, the solutions are $n = 1, 2$.

4. Note that $n = 1, 2$ are not solutions. Suppose $n \geqslant 3$. If $d(n) = n - 1$, then exactly one of $1, 2, \ldots, n$ is not a divisor of n. In particular, we have $n - 1 \mid n$ or $n - 2 \mid n$. The former case yields $n - 1 \mid 1$, which has no solution as $n \geqslant 3$. The latter case yields $n - 2 \mid 2$. The only solutions are $n = 3, 4$.

   We check that $d(3) = 2 = 3 - 1$ and $d(4) = 3 = 4 - 1$. Therefore, the solutions are $n = 3, 4$.

5. Clearly, $n = 1$ is a solution. For $n \geqslant 2$, let $n = p_1^{a_1} p_2^{a_2} \cdots p_t^{a_t}$ be the prime factorization of n. Then
$$d(n) = (a_1 + 1)(a_2 + 1) \cdots (a_t + 1).$$
   If $d(n)$ is odd, then each $a_j + 1$ is odd. This means each a_j is even. It follows that
$$n = (p_1^{\frac{a_1}{2}} p_2^{\frac{a_2}{2}} \cdots p_t^{\frac{a_t}{2}})^2$$
   is a perfect square. As the steps are reversible, we find that n can be any perfect square (including 1).

6. For example, we may take $n = p^{m-1}$ for some prime p. Then $d(n) = d(p^{m-1}) = m$. Clearly, there are infinitely many choices of n.

7. For example, we may take $m = 3^k$ and $n = 3^{k+1}$ so that
$$\omega(m + n) = \omega(4 \times 3^k) = 2$$
   and
$$\omega(m) + \omega(n) = \omega(3^k) + \omega(3^{k+1}) = 1 + 1 = 2 = \omega(m + n).$$

8. If $n = 2^{p-1}(2^p - 1)$ for some prime $2^p - 1$, then

$$\sigma(n) = (1 + 2 + 2^2 + \cdots + 2^{p-1})(1 + (2^p - 1)) = (2^p - 1)2^p = 2n.$$

Conversely, suppose $\sigma(n) = 2n$. Let $n = 2^{p-1}m$ for some integer $p \geqslant 2$ (as n is even) and odd m. Then

$$\sigma(n) = (1 + 2 + 2^2 + \cdots + 2^{p-1})\sigma(m) = (2^p - 1)\sigma(m)$$

since the sum of divisors function is multiplicative. This implies

$$2^p m = (2^p - 1)\sigma(m).$$

As $(2^p, 2^p - 1) = 1$, we must have $2^p - 1 \mid m$. Let $m = (2^p - 1)k$. Then the equation becomes

$$2^p k = \sigma((2^p - 1)k).$$

Note that k and $(2^p - 1)k$ are distinct positive divisors of $(2^p - 1)k$. As their sum is $2^p k$, these are the only positive divisors of $(2^p - 1)k$. In particular, we have $k = 1$ and $2^p - 1$ is a prime. Hence, $n = 2^{p-1}(2^p - 1)$.

So the equivalence is established. Note that one can also show that p must be a prime.

<u>Euler's Totient Function</u>

9. Since $15 = 3 \times 5$ and $40 = 2^3 \times 5$, we have

$$\begin{aligned} \varphi(15) &= 15\Big(1 - \frac{1}{3}\Big)\Big(1 - \frac{1}{5}\Big) = 8, \\ \varphi(40) &= 40\Big(1 - \frac{1}{2}\Big)\Big(1 - \frac{1}{5}\Big) = 16. \end{aligned}$$

10. For each integer n such that $n \equiv r \pmod{28}$, we have $(n, 28) = (r, 28)$. Therefore, the number of integers from $28k + 1$ to $28(k + 1)$ which are relatively prime to 28 is $\varphi(28)$ for each $k = 0, 1, \ldots, 9$. Thus, the answer is

$$10\varphi(28) = 10 \cdot 28\Big(1 - \frac{1}{2}\Big)\Big(1 - \frac{1}{7}\Big) = 120.$$

11. Since the Euler's totient function is multiplicative, it suffices to show that $\varphi(p)$ and $\varphi(2^k)$ are even for any prime $p \geqslant 3$ and integer $k \geqslant 2$. This is true since $\varphi(p) = p - 1$ and $\varphi(2^k) = 2^{k-1}$ are even.

12. Firstly, $n = 1$ is a possibility as $\varphi(1) = 1$. For $n \geqslant 2$, let $n = p_1^{a_1} p_2^{a_2} \cdots p_t^{a_t}$ be the prime factorization of n.

    If $8 \mid n$, we may assume $p_1 = 2$ and $a_1 \geqslant 3$. Note that $\varphi(2^{a_1}) \mid \varphi(n)$ from the multiplicative property of $\varphi(n)$. As $\varphi(2^{a_1}) = 2^{a_1-1}$, we find that $4 \mid \varphi(n)$.

    If $4 \parallel n$, we claim that $\varphi(n)$ is divisible by 4 unless $n = 4$. The case $n = 4$ is trivial since $\varphi(4) = 2$. If $t \geqslant 2$, then $\varphi(4)\varphi(p_2^{a_2}) \mid \varphi(n)$. As $\varphi(4) = 2$ and $\varphi(p_2^{a_2}) = p_2^{a_2-1}(p_2 - 1)$ are even, we have $4 \mid \varphi(n)$.

    If $2 \parallel n$, then $\varphi(n) = \varphi(2)\varphi\Big(\dfrac{n}{2}\Big) = \varphi\Big(\dfrac{n}{2}\Big)$. Thus, this reduces to the remaining cases (and the trivial case $\varphi(2) = 1$).

    If $2 \nmid n$ and $t \geqslant 2$, then $\varphi(p_1^{a_1})\varphi(p_2^{a_2}) \mid \varphi(n)$. As the numbers $\varphi(p_1^{a_1}) = p_1^{a_1-1}(p_1 - 1)$ and $\varphi(p_2^{a_2}) = p_2^{a_2-1}(p_2 - 1)$ are even, we have $4 \mid \varphi(n)$.

    If $2 \nmid n$ and $t = 1$, then $\varphi(n) = \varphi(p_1^{a_1}) = p_1^{a_1-1}(p_1 - 1)$. This is divisible by 4 if and only if $p_1 \equiv 1 \pmod 4$.

    It follows that $4 \nmid \varphi(n)$ if $n = 1, 2, 4$ or $n = p^a, 2p^a$ for some prime $p = 4k + 3$ and some positive integer a.

13. Obviously, we have $n \geqslant 2$. Let $n = p_1^{a_1} p_2^{a_2} \cdots p_t^{a_t}$ be the prime factorization of n. Then

    $$n\Big(1 - \frac{1}{p_1}\Big)\Big(1 - \frac{1}{p_2}\Big) \cdots \Big(1 - \frac{1}{p_t}\Big) = 2,$$

    i.e.

    $$p_1^{a_1-1} p_2^{a_2-1} \cdots p_t^{a_t-1}(p_1 - 1)(p_2 - 1) \cdots (p_t - 1) = 2.$$

    If $t = 1$, then we need $p_1^{a_1-1}(p_1 - 1) = 2$. Thus, $p_1 - 1$ can only be 1 or 2. If $p_1 = 2$, then we have $a_1 = 2$. This gives a solution $\varphi(4) = 2$. If $p_1 = 3$, then we have $a_1 = 1$. This gives a solution $\varphi(3) = 2$.

    If $t = 2$, then we need $p_1^{a_1-1} p_2^{a_2-1}(p_1 - 1)(p_2 - 1) = 2$. Note that

    $$(p_1 - 1)(p_2 - 1) \geqslant (2 - 1)(3 - 1) = 2.$$

    Thus, equality holds and $a_1 = a_2 = 1$. This gives a solution $\varphi(6) = 2$.

    If $t \geqslant 3$, then $(p_1 - 1)(p_2 - 1) \cdots (p_t - 1) \geqslant (2 - 1)(3 - 1)(5 - 1) > 2$. There is no solution.

    Therefore, the solutions are $n = 3, 4, 6$.

14. Obviously, we have $n \geqslant 2$. Let $n = p_1^{a_1} p_2^{a_2} \cdots p_t^{a_t}$ be the prime factorization of n. Then

    $$n\Big(1 - \frac{1}{p_1}\Big)\Big(1 - \frac{1}{p_2}\Big) \cdots \Big(1 - \frac{1}{p_t}\Big) = 4,$$

    i.e.

    $$p_1^{a_1-1} p_2^{a_2-1} \cdots p_t^{a_t-1}(p_1 - 1)(p_2 - 1) \cdots (p_t - 1) = 4.$$

If $t = 1$, then we need $p_1^{a_1-1}(p_1 - 1) = 4$. Thus, $p_1 - 1$ can only be $1, 2, 4$. If $p_1 = 2$, then we have $a_1 = 3$. This gives a solution $\varphi(8) = 4$. If $p_1 = 3$, there is no solution. If $p_1 = 5$, then we have $a_1 = 1$. This gives a solution $\varphi(5) = 4$.

If $t = 2$, then we need $p_1^{a_1-1}p_2^{a_2-1}(p_1 - 1)(p_2 - 1) = 4$. If $(p_1, p_2) = (2, 3)$, then we have $2^{a_1-1}3^{a_2-1} = 2$. We get $a_1 = 2$ and $a_2 = 1$. This gives a solution $\varphi(12) = 4$. If $(p_1, p_2) = (2, 5)$, then $a_1 = a_2 = 1$. This gives a solution $\varphi(10) = 4$. In the remaining cases, we have $(p_1-1)(p_2-1) \geqslant (2-1)(7-1) > 4$ or $(p_1-1)(p_2-1) \geqslant (3-1)(5-1) > 4$. There is no solution.

If $t \geqslant 3$, then $(p_1 - 1)(p_2 - 1)\cdots(p_t - 1) \geqslant (2 - 1)(3 - 1)(5 - 1) > 4$. There is no solution.

Therefore, the solutions are $n = 5, 8, 10, 12$.

15. It suffices to consider $n \geqslant 2$. Let $n = p_1^{a_1}p_2^{a_2}\cdots p_t^{a_t}$ be the prime factorization of n. Then

$$n\Big(1 - \frac{1}{p_1}\Big)\Big(1 - \frac{1}{p_2}\Big)\cdots\Big(1 - \frac{1}{p_t}\Big) = m,$$

i.e.

$$p_1^{a_1-1}p_2^{a_2-1}\cdots p_t^{a_t-1}(p_1 - 1)(p_2 - 1)\cdots(p_t - 1) = m.$$

Since $p_j - 1 \leqslant m$ and m is fixed, each p_j is bounded above. Also, since $p_j^{a_j-1} \leqslant m$, each a_j is bounded above. In addition, we have

$$m \geqslant (p_1 - 1)(p_2 - 1)\cdots(p_t - 1) \geqslant (2 - 1)(3 - 1)^{t-1} = 2^{t-1}.$$

This shows t is bounded. As every variable is bounded, there are finitely many solutions.

16. The case $n = 1$ is trivial. For $n \geqslant 2$, let $n = p_1^{a_1}p_2^{a_2}\cdots p_t^{a_t}$ be the prime factorization of n. Then

$$\begin{aligned}\sum_{d|n}\varphi(d) &= \sum_{\substack{b_1,\ldots,b_t\\ 0\leqslant b_j\leqslant a_j}} \varphi(p_1^{b_1}p_2^{b_2}\cdots p_t^{b_t})\\ &= \sum_{\substack{b_1,\ldots,b_t\\ 0\leqslant b_j\leqslant a_j}} \varphi(p_1^{b_1})\varphi(p_2^{b_2})\cdots\varphi(p_t^{b_t})\\ &= \Big(\sum_{0\leqslant b_1\leqslant a_1}\varphi(p_1^{b_1})\Big)\Big(\sum_{0\leqslant b_2\leqslant a_2}\varphi(p_2^{b_2})\Big)\cdots\Big(\sum_{0\leqslant b_t\leqslant a_t}\varphi(p_t^{b_t})\Big).\end{aligned}$$

Note that

$$\sum_{0\leqslant b_j\leqslant a_j}\varphi(p_j^{b_j}) = 1 + \sum_{1\leqslant b_j\leqslant a_j} p_j^{b_j-1}(p_j - 1) = 1 + (p_j - 1)\cdot\frac{p_j^{a_j} - 1}{p_j - 1} = p_j^{a_j}.$$

Therefore, we have

$$\sum_{d|n}\varphi(d) = p_1^{a_1}p_2^{a_2}\cdots p_t^{a_t} = n.$$

17. The case $m = 1$ or $n = 1$ is trivial. For $m, n \geqslant 2$, we let $m = p_1^{a_1} p_2^{a_2} \cdots p_t^{a_t}$ and $n = p_1^{b_1} p_2^{b_2} \cdots p_t^{b_t}$ be the prime factorizations of m and n. Firstly, as $\varphi(a)$ is multiplicative, we have

$$\varphi(mn) = \varphi\Big(\prod_{j=1}^{t} p_j^{a_j+b_j}\Big) = \prod_{j=1}^{t} \varphi(p_j^{a_j+b_j}) = \prod_{j=1}^{t} p_j^{a_j+b_j-1}(p_j - 1).$$

Next, we have $d = p_1^{m_1} p_2^{m_2} \cdots p_t^{m_t}$ where $m_j = \min\{a_j, b_j\}$ for each j. We find that

$$\begin{aligned}
\varphi(m)\varphi(n)\frac{d}{\varphi(d)} &= \varphi\Big(\prod_{j=1}^{t} p_j^{a_j}\Big)\varphi\Big(\prod_{j=1}^{t} p_j^{b_j}\Big)\Big(\prod_{j=1}^{t} p_j^{m_j}\Big)\varphi\Big(\prod_{j=1}^{t} p_j^{m_j}\Big)^{-1} \\
&= \prod_{j=1}^{t} \varphi(p_j^{a_j})\varphi(p_j^{b_j})(p_j^{m_j})\varphi(p_j^{m_j})^{-1} \\
&= \prod_{j=1}^{t} p_j^{a_j-1}(p_j - 1)p_j^{b_j-1}(p_j - 1)p_j^{m_j} p_j^{-(m_j-1)}(p_j - 1)^{-1} \\
&= \prod_{j=1}^{t} p_j^{a_j+b_j-1}(p_j - 1) = \varphi(mn).
\end{aligned}$$

18. Let $d = (m, n)$. Then the equation becomes

$$\varphi(m)\varphi(n)\frac{d}{\varphi(d)} = \varphi(m) + \varphi(n),$$

i.e.

$$\frac{d}{\varphi(d)} = \frac{1}{\varphi(n)} + \frac{1}{\varphi(m)}.$$

Note that the left-hand side is not less than 1. If $\varphi(n), \varphi(m) \geqslant 2$, then we must have $\varphi(n) = \varphi(m) = 2$ and $d = \varphi(d)$. The only positive integer d satisfying $\varphi(d) = d$ is $d = 1$. From $\varphi(n) = \varphi(m) = 2$, the integers m, n can be $3, 4, 6$. Since $(m, n) = 1$, the solutions are $(m, n) = (3, 4), (4, 3)$.

If $\varphi(n) = 1$ (or $\varphi(m) = 1$), then $n = 1$. The original equation becomes $\varphi(m) = \varphi(m)+1$, which has no solution.

Therefore, the only solutions are $(m, n) = (3, 4), (4, 3)$.

Floor Function

19. If $x < 0$, then $\dfrac{10}{x}, \dfrac{x}{10} < 0$. The left-hand side is negative.

If $x \geqslant 0$, then the two terms on the left are nonnegative integers. This gives

$$\left\lfloor \frac{10}{x} \right\rfloor = 0, \ \left\lfloor \frac{x}{10} \right\rfloor = 1$$

or

$$\left\lfloor \frac{10}{x} \right\rfloor = 1, \ \left\lfloor \frac{x}{10} \right\rfloor = 0.$$

For the former case, we need $\frac{10}{x} < 1$ and $1 \leqslant \frac{x}{10} < 2$. This holds if and only if $10 < x < 20$.

For the latter case, we need $1 \leqslant \frac{10}{x} < 2$ and $\frac{x}{10} < 1$. This holds if and only if $5 < x < 10$.

Therefore, x can be any real numbers satisfying $5 < x < 20$ except $x = 10$.

20. Let $n = a^2 + b$ where $0 \leqslant b < 2a + 1$. Then we need to prove

$$\left\lfloor \frac{a^2 + b}{a + 1} \right\rfloor = a \quad \text{or} \quad a - 1.$$

Indeed, if $0 \leqslant b < a$, then $a - 1 < \frac{a^2 + b}{a + 1} < a$ so that the left-hand side is equal to $a - 1$.

If $a \leqslant b < 2a + 1$, then $a \leqslant \frac{a^2 + b}{a + 1} < a + 1$ so that the left-hand side is equal to a.

21. Let $n = a^2 + b$ where $0 \leqslant b < 2a + 1$. Observe the following pattern:

$$\begin{aligned} \lfloor\sqrt{1}\rfloor = \lfloor\sqrt{2}\rfloor = \lfloor\sqrt{3}\rfloor &= 1, \\ \lfloor\sqrt{4}\rfloor = \lfloor\sqrt{5}\rfloor = \cdots = \lfloor\sqrt{8}\rfloor &= 2, \\ \lfloor\sqrt{9}\rfloor = \lfloor\sqrt{10}\rfloor = \cdots = \lfloor\sqrt{15}\rfloor &= 3, \\ &\cdots \\ \lfloor\sqrt{(a-1)^2}\rfloor = \lfloor\sqrt{(a-1)^2 + 1}\rfloor = \cdots = \lfloor\sqrt{a^2 - 1}\rfloor &= a - 1, \\ \lfloor\sqrt{a^2}\rfloor = \lfloor\sqrt{a^2 + 1}\rfloor = \cdots = \lfloor\sqrt{a^2 + b}\rfloor &= a. \end{aligned}$$

For each $k = 1, 2, \ldots, a - 1$, there are exactly $2k + 1$ terms equal to k. Therefore, the sum is equal to

$$\begin{aligned} \sum_{k=1}^{a-1} k(2k+1) + a(b+1) &= 2\sum_{k=1}^{a-1} k^2 + \sum_{k=1}^{a-1} k + a(b+1) \\ &= 2 \cdot \frac{(a-1)a(2a-1)}{6} + \frac{(a-1)a}{2} + a(n - a^2 + 1) \\ &= an - \frac{a(a-1)(2a+5)}{6}. \end{aligned}$$

22. Note that

$$\left\lfloor \frac{33k}{100} \right\rfloor = \frac{33k}{100} - \left\{ \frac{33k}{100} \right\}.$$

As $(33, 100) = 1$, the term $33k$ runs through all nonzero congruence classes modulo 100 when k runs from 1 to 99. Thus, the sum is equal to

$$\begin{aligned} \sum_{k=1}^{99} \frac{33k}{100} - \sum_{k=1}^{99} \left\{ \frac{33k}{100} \right\} &= \frac{33}{100} \sum_{k=1}^{99} k - \sum_{k=1}^{99} \frac{k}{100} \\ &= \left(\frac{33}{100} - \frac{1}{100} \right) \cdot \frac{99 \cdot 100}{2} = 1584. \end{aligned}$$

23. Let $n = 2^{a_1} + 2^{a_2} + \cdots + 2^{a_d}$ where $0 \leqslant a_1 < a_2 < \cdots < a_d$. This is the same as writing n in binary representation.

For each $j \geqslant 0$, if $j = a_i$ for some i, then

$$\begin{aligned} \left\lfloor \frac{n + 2^j}{2^{j+1}} \right\rfloor &= \left\lfloor \frac{2^{a_1} + \cdots + 2^{a_{i-1}} + 2^{a_i+1} + 2^{a_{i+1}} + \cdots + 2^{a_d}}{2^{a_i+1}} \right\rfloor \\ &= \left\lfloor \frac{2^{a_i+1} + 2^{a_{i+1}} + \cdots + 2^{a_d}}{2^{a_i+1}} \right\rfloor = \left\lfloor \frac{n}{2^{a_i+1}} \right\rfloor + 1. \end{aligned}$$

If $j \neq a_i$ for all i, then there is no carry to the exponents. Thus,

$$\left\lfloor \frac{n + 2^j}{2^{j+1}} \right\rfloor = \left\lfloor \frac{n}{2^{j+1}} \right\rfloor.$$

Summing over all j's, the desired quantity S becomes

$$d + \left\lfloor \frac{n}{2^1} \right\rfloor + \left\lfloor \frac{n}{2^2} \right\rfloor + \cdots.$$

It is not hard to see that

$$\left\lfloor \frac{2^{a_1} + \cdots + 2^{a_d}}{2^j} \right\rfloor = \left\lfloor \frac{2^{a_1}}{2^j} \right\rfloor + \cdots + \left\lfloor \frac{2^{a_d}}{2^j} \right\rfloor$$

for any j as the exponents are distinct. This yields

$$\begin{aligned} S = d + \sum_{j=1}^{\infty} \left\lfloor \frac{n}{2^j} \right\rfloor &= d + \sum_{j=1}^{\infty} \sum_{i=1}^{d} \left\lfloor \frac{2^{a_i}}{2^j} \right\rfloor = d + \sum_{i=1}^{d} \sum_{j=1}^{\infty} \left\lfloor \frac{2^{a_i}}{2^j} \right\rfloor \\ &= d + \sum_{i=1}^{d} \sum_{j=1}^{a_i} \frac{2^{a_i}}{2^j} = d + \sum_{i=1}^{d} (2^{a_i} - 1) = n. \end{aligned}$$

24. Suppose on the contrary that $2^k < n < 2^{k+1}$ for some $k \geqslant 2$. Then we have

$$\left\lfloor \frac{2^n}{n} \right\rfloor \leqslant \frac{2^n}{n} < 2^{n-k}$$

and

$$\left\lfloor \frac{2^n}{n} \right\rfloor > \frac{2^n}{n} - 1 > \frac{2^n}{2^{k+1}-1} - 1.$$

From the assumption, we must have $\left\lfloor \frac{2^n}{n} \right\rfloor \leqslant 2^{n-k-1}$. This yields

$$2^{n-k-1} > \frac{2^n}{2^{k+1}-1} - 1,$$

i.e. $2^n - 2^{n-k-1} > 2^n - (2^{k+1} - 1)$. This implies

$$2^{k+1} > 2^{n-k-1} + 1,$$

and hence $k + 1 > n - k - 1$. From $n > 2^k$, we get

$$2k + 2 > n > 2^k.$$

By induction, it is easy to prove that $2k + 2 \leqslant 2^k$ for $k \geqslant 3$. Hence, we must have $k = 2$. It suffices to check $n = 5, 6, 7$. However, none of these makes $\left\lfloor \frac{2^n}{n} \right\rfloor$ a power of 2. Therefore, n is a power of 2.

25. Suppose on the contrary that $P \neq Q$. Then the polynomial $P - Q$ has finitely many real roots. Therefore, $P(x) - Q(x) \neq 0$ for sufficiently large x. WLOG we may assume $P(x) > Q(x)$ for all sufficiently large x (note that we cannot have $P(a) > Q(a)$ and $P(b) < Q(b)$ for large a, b by the intermediate value theorem).

As $P(x)$ is a non-constant polynomial, there exists an integer n (positive or not) and a large real number y such that $P(y) = n$. In that case, we have

$$\lfloor Q(y) \rfloor = \lfloor P(y) \rfloor = n.$$

This yields $Q(y) \geqslant n = P(y)$, which contradicts $P(x) > Q(x)$.

Therefore, we must have $P = Q$.